

CMP-02 - Identity and Continuity Service

Maintain durable governed identity and continuity for HAL, humans, services, devices, nodes, and Presences.

Authority and state ownership

Book I: Decisions 1-7, 24-29, 35, 40, 47-50, and 58. Book II: Chapters 04, 28. Authoritative state: Identity Records, identifier namespaces, continuity lineage, identity lifecycle, authentication-assurance bindings. Explicitly non-owned: Owner authority, delegations, trust scores, credentials or secret material.

Critical invariant

One constitutional HAL identity persists across all Presences, runtimes, models, services, and recovery events.

Logical interfaces

CMP-02-IF-01 / Command / Create Identity

Command `Create Identity` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-02 / Command / Bind Identifier

Command `Bind Identifier` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-03 / Command / Suspend Identity

Command `Suspend Identity` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-04 / Command / Reconcile Continuity

Command `Reconcile Continuity` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-05 / Query / Resolve Identity

Query `Resolve Identity` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-06 / Query / Get Continuity Lineage

Query `Get Continuity Lineage` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-07 / Query / Get Authentication Context

Query `Get Authentication Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-08 / Event / Identity Created

Event `Identity Created` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-09 / Event / Identity Suspended

Event `Identity Suspended` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-02-IF-10 / Event / Continuity Reconciled

Event `Continuity Reconciled` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Proposed > Active > Suspended > Retired; continuity: Unverified > Verified > Disputed. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: duplicate HAL identity, identifier collision, lineage break, stale authentication binding, identity-source disagreement. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-02-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-02-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-02-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-02-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-02-TST-005 - Critical invariant. One constitutional HAL identity persists across all Presences, runtimes, models, services, and recovery events. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-02-TST-006 - Failure containment. Inject duplicate HAL identity, identifier collision, lineage break, stale authentication binding, identity-source disagreement. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-02-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-02-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-02-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-02-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.